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What Did She Know About Transformation That We Don't?

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Abstract

An old woman lingers in Sir Bertilak's castle, silent and unnoticed. Only at the end of *Sir Gawain and the Green Knight* is her name revealed—Morgan le Fay. A fleeting presence, a shadow behind the game. But was she merely a sorceress? Or a master of deeper knowledge, a scientist of transformation cloaked in mysticism? This paper reimagines Morgan le Fay through the lens of *Le Morte d'Arthur*, *Vita Merlini*, and medieval alchemical-herbal traditions. Far from a mere enchantress, her manipulation of life forms, elemental forces, and bodily change aligns with alchemy's quest for transmutation, renewal and modern-day Pharmacology. Her creation of the Green Knight is no illusion—it echoes the origins of genetic manipulation. In medieval thought, metals were purified into gold through trial, just as Bertilak becomes a vessel of endurance and near-immortality to test Gawain's virtue. The Green Knight's seasonal return and survival of decapitation embody alchemical ideals of regeneration and the Elixir of Life. Could Morgan's "sorcery" have roots in early proto-scientific knowledge? This interpretation is grounded in *The Mirror of Alchemy* (Roger Bacon), *The Book of Secrets* (Pseudo-Albertus Magnus), and *The Emerald Tablet* (Hermes Trismegistus), texts that blur the natural and the unnatural. Morgan's imagined experimentations bridge medieval alchemy and modern debates on bioengineering, human enhancement, and life's ethical boundaries. Perhaps Morgan wasn't just shaping legend, but prefiguring the science and the dilemmas of the future.

Keywords: Morgan le Fay, Medieval alchemy, Bioengineering, Pharmacology, Sorcery and science, Renaissance Women.

INTRODUCTION

She appears without ceremony. Seated beside a noblewoman, aged and overlooked, her presence is brushed aside as merely atmospheric. In *Sir Gawain and the Green Knight*, she is introduced like an afterthought—unnamed, undisturbed (Tolkien, 1975, lines 2411–2460). Until, at the very end, her name surfaces. A name that should have commanded attention. Instead, it barely registers. Readers accept her as the “enchantress,” the plot’s puppet-master, and move on, satisfied.

But should we be?

This paper refuses to move on.

In the folds of medieval romance, certain female figures carry the weight of forgotten knowledge, encoded in myth, buried under centuries of interpretive neglect (Malory, 1998; Chrétien de Troyes, 1991). This unnamed woman, later identified as Morgan le Fay, is one such figure. Often flattened into trope—the witch, the crone, the antagonist—she is rarely studied as a thinker in her own right. Yet the more one examines her presence across medieval texts, the more a different narrative begins to flicker through the cracks. Beneath the veil of enchantment lies something richer: a deep, embodied knowledge of transformation, healing, and matter itself.

This paper reconsiders Morgan le Fay not as a fantastical outlier but as a figure whose so-called magic aligns with early medical and alchemical traditions. Her association with herbs, bodily metamorphosis, and the manipulation of nature echoes the practices of women working in the blurred boundaries between science and sorcery, where empirical experimentation was veiled

as magic to survive male suspicion. Morgan is not merely a literary device; she is a symbol of how feminine knowledge has historically been rendered illegible (Barefield, 2000).

To read Morgan seriously is to ask uncomfortable questions: What knowledge systems are allowed to be called science? Who gets to be a philosopher, a healer, a theorist? And why, even now, do women who transgress boundaries still get branded as witches?

By tracing Morgan le Fay's evolution across texts and interpretations, this study reveals the historical processes that transformed women's expertise into myth, their agency into danger, and their intellect into silence. In that silence, however, something powerful lingers. This is not just a recovery—it is a reckoning.

Let us ask the question that slips through the cracks of both legend and scholarship—and dare to find the answer no one thought to seek: What did she know about transformation that we don't?

This paper unfolds in three parts. First, it reads the Green Knight as Morgan le Fay's alchemical experiment. Next, it explores her shape-shifting as proto-bioengineering. Finally, it reimagines her as a veiled scientist, centuries ahead of her time. Let's begin with the alchemy.

1.Morgan's Magnum Opus: The Green Knight as an Alchemical Experiment

1.1 The Philosophical and Spiritual Dimensions of Alchemy

The long-standing perception of medieval alchemy as a mere pursuit of material wealth through the creation of gold is a reductive oversimplification. The true essence of alchemy, as outlined by Roger Bacon in *The Mirror of Alchemy*, extends far beyond the transformation of base

metals. Alchemy, according to Bacon, was “a science teaching how to make and compound a certain medicine, which is called Elixir, the which, when it is cast upon metals or imperfect bodies, doth fully perfect them in the very projection” (Bacon, 2018, p. 2). This “Elixir” is not just a substance of material transformation but a vehicle for spiritual and philosophical refinement. The goal of alchemy was not only to transmute metals but to perfect the alchemist’s soul, elevating them to a state of transcendence.

Bacon’s assertion that “the whole body [is] a spiritual, fixed thing” (p. 4) encapsulates the dual nature of alchemical work: it is a pursuit of both material perfection and spiritual enlightenment. The alchemist is not just concerned with external change but with an internal process of purification and elevation. Similarly, the Hermetic tradition, represented in the *Smargadine Table*, emphasizes the unity of the physical and the spiritual, stating, “That which is beneath is like that which is above, and that which is above is like that which is beneath, to work the miracles of one thing” (The Emerald Tablet, n.d., para. 2). This mirrored relationship between the macrocosm (the universe) and the microcosm (the individual) underlines the alchemical idea that material transmutation mirrors the spiritual refinement of the soul.

Alchemy, as seen through the lens of Hermes Trismegistus, also speaks to the dual nature of existence, dividing the Philosopher’s Stone into two parts: “the inferior part, which remaineth beneath” (the body) and “the superior part, which is the Soul, which quickeneth the whole Stone and raiseth it up” (Magnus, n.d., p. 18). In this framework, alchemy is a path that leads to both physical and spiritual immortality, guiding the individual toward a state of enlightened perfection. This philosophical foundation illuminates the deeper purpose of alchemy: to achieve not just material wealth but spiritual enlightenment and resilience.

1.2 The Green Knight: An Alchemical Figure

The *Green Knight* in *Sir Gawain and the Green Knight* can be read as a physical manifestation of these alchemical principles. Upon his entrance to Arthur's court, the Green Knight is an otherworldly figure: "All of the green were they made, both garments and man," riding "a green horse great and thick," with a "holly-bundle... greatest in greenery when groves are leafless" (Tolkien, 1975, Line 210). The vivid green imagery here is not coincidental; it is an alchemical symbol of *viriditas*, the raw potential of life. The Green Knight's very appearance symbolises the unrefined, primal state of nature before undergoing the transformative process.

His challenge to Gawain—to strike him with his axe and receive a return blow in a year and a day—can be interpreted as an invitation to undergo an alchemical process. When Gawain decapitates him, the Green Knight's miraculous survival—he picks up his severed head and rides away—serves as a metaphor for the alchemical *nigredo*, the first stage of the Great Work. In alchemical terms, *nigredo* represents the "blackening" or putrefaction, where the substance is broken down, just as Gawain's initial action appears to bring destruction. However, rather than death, this stage of disintegration is a necessary step for transformation, as "in that blackness, whiteness is hidden" (Bacon, 2018, p. 9).

The Green Knight's resilience in the face of death signifies a triumph over mortality, a key theme in alchemy's quest for immortality and the Elixir of Life.

1.3 Gawain's Journey: An Alchemical Metaphor

Gawain's journey throughout the text mirrors the stages of alchemical transformation, which can be divided into three distinct phases: **nigredo**, **albedo**, and **rubedo**. Each of these stages

corresponds to Gawain's trials and moral evolution.

- **Nigredo (Blackening):** The beginning of Gawain's journey—his harsh winter journey through the wilderness—symbolizes the breakdown of the self, a process akin to *nigredo* in alchemy. This is the stage where Gawain is stripped of his pretensions and forced to confront his vulnerability, symbolizing the dissolution of the ego in alchemical thought.
- **Albedo (Whitening):** Gawain's trials with Lady Bertilak represent the *albedo* phase, wherein Gawain must demonstrate purity of spirit. His restraint, honesty, and adherence to his code of honor are tests of his inner purification. The whiteness symbolized by the *albedo* stage corresponds to the purification of the soul as Gawain moves closer to spiritual clarity.
- **Rubedo (Reddening):** Finally, Gawain's return and his acceptance of the return blow represent *rubedo*, the final stage of the alchemical process. This stage is marked by the reddening of the substance, symbolizing the full integration of the self and the attainment of perfection. Gawain's recognition of his flaws and his willingness to accept the consequences of his actions signify his completion of the alchemical transformation.

Through this journey, Gawain transcends his original, imperfect state, emerging as a more enlightened individual. This mirrors the alchemical pursuit of turning base metals into gold, not just in a literal sense, but in the sense of achieving spiritual refinement and enlightenment.

1.4 The Alchemical Palette: The Secret of Colours

One of the most fascinating aspects of alchemical transformation is the symbolic significance of color. In the alchemical tradition, different colors emerge during the purification of base

materials into gold. These colors—green, red, and gold—are central to the process of transformation, and they appear with striking frequency in *Sir Gawain and the Green Knight*.

- **Green:** In alchemical texts, green symbolizes *viriditas*, the raw potential for life and growth. As Bacon states, “Concoct it till it appears green unto thee, and that is the soul” (Bacon, 2018, p. 10). The Green Knight himself embodies this stage, representing both the unrefined material and the primal force of nature waiting for transformation.
- **Red:** The color red signifies the final stage of purification in alchemy—*rubedo*. “It maketh itself red” (Bacon, 2018, p. 12) as the substance reaches its highest state of refinement. Gawain’s trial by blood, symbolized by the wound he receives from the Green Knight, marks this stage in his personal transformation.
- **Gold:** Gold, the ultimate goal of alchemy, symbolizes the perfected state of both matter and spirit. Bacon describes the red Elixir as having the power to “change all metals into pure gold” (p. 16). Gawain’s completion of his trial and his ultimate enlightenment reflect this final stage of refinement, where he attains a state of spiritual and moral purity.

These colors are not just decorative but serve as a coded language that aligns with alchemical principles. The Green Knight’s challenge to Gawain is an invitation to undergo the same transformative process, beginning with the raw green potential and culminating in the red and golden stages of spiritual perfection.

1.5 *Morgan le Fay as the Alchemist*

If the Green Knight’s challenge and Gawain’s subsequent trials mirror an alchemical experiment, then Morgan le Fay must be seen as the alchemist orchestrating the transformation.

The text makes this explicit when it reveals that the Green Knight is “enchanted and changed in my hue” by Morgan’s “cunning of lore and crafts well learned” (Tolkien, 1975, line 2446). Through this magical transformation, Morgan le Fay initiates an alchemical process that purifies Gawain, both physically and spiritually. Gawain, in turn, is “as good was acknowledged and as gold refined” (Tolkien, 1975, line 2446), signifying that the trial has achieved its intended result: the transmutation of base character into moral gold.

Thus, Morgan le Fay’s magic can be understood not merely as sorcery but as a form of proto-scientific bioengineering, aligning her powers with the alchemical pursuit of perfection. Her mastery of transformation is not just a fantastical concept but a reflection of the deeply philosophical and spiritual processes found in the alchemical tradition.

2. Reimagining the Magic: The Origins of Genetic Manipulation

2.1 The "Man to Stone" Transformation: A Modern Scientific Perspective

In *Le Morte d'Arthur*, Morgan le Fay’s ability to turn herself and her horse into stone is one of the most striking examples of magical shape-shifting. This transformation goes beyond simple illusion, suggesting a profound alteration of both physical form and material essence:

Then she rode into a valley where many great stones were, and when she saw she must be overtaken, she shaped herself, horse and man, by enchantment unto a great marble stone. So when Arthur was gone she turned all into the likeliness as she and they were before, and said, 'Sirs, now may we go where we will.' (Malory, 1998, p. 312)

At a glance, this magical transformation appears fantastical. However, from the lens of modern science, it raises intriguing questions about material manipulation and the theoretical

possibilities of altering the structure of matter.

2.2 Shape-Shifting and Material Science: Understanding the Impossible

Modern scientific advancements, particularly in material science and bioengineering, provide an intriguing framework to understand how a transformation akin to Morgan's might be conceptualized today. Though the medieval imagination could not have known the specifics, today we are learning how to manipulate the material properties of substances at the atomic and molecular levels. The idea of turning something biological, like a human or animal, into something as inorganic as stone, is not entirely beyond the realm of speculative science.

One relevant field to this discussion is **totimorphic materials**—materials that can change their shape in response to external stimuli, such as temperature, light, or pressure. Recent research in the field of **shape-shifting materials** is opening up the possibility of creating structures that can alter their form, not just in response to external forces but through intrinsic changes at the molecular level (Burrows, 2021).

For instance, scientists have developed **self-healing materials** that can "repair" themselves after damage, inspired by the regenerative qualities of biological systems. While these materials do not turn into stone per se, they are capable of mimicking certain "magical" qualities, such as regrowth or reconfiguration, that seem to echo Morgan's mystical abilities. This work shows that while total transformations from flesh to stone remain purely in the realm of fantasy, the manipulation of materials at the cellular or molecular level is very real.

2.3 Bioengineering and Molecular Biology: The Science of Transformation

Morgan's transformation of herself and her horse into stone could also be examined through the lens of **bioengineering**. In bioengineering, the genetic code is manipulated to alter the characteristics of an organism, making it stronger, more resilient, or capable of new functions (Centers for Disease Control and Prevention, n.d.). This aligns with the idea that Morgan's transformation might have involved changing the molecular structure of her and her horse, reshaping them into solid forms at a microscopic level.

In the context of modern biology, there is ongoing research into the manipulation of **biomaterials**—organic substances derived from living organisms, which can be altered to possess unique properties. This type of research could one day lead to the creation of biological entities capable of withstanding extreme conditions, such as turning into an inorganic state (like stone), or other radical transformations in response to environmental cues.

Researchers are particularly interested in the regenerative capabilities of plant-based materials. For example, **regenerative medicine** has made strides in developing materials that mimic the regenerative abilities of trees and other plants, where tissues or organs could be “healed” or “regrown” after damage (Marais et al., 2018). If we follow this logic, it's possible to imagine a future where bioengineered organisms could regenerate parts of their bodies or change their material properties in response to environmental signals, much like the way Morgan's enchantment would cause her body to revert from stone to flesh.

2.4 Molecular Shape-Shifting: Quantum Biology's Role in Transformation

Another layer to Morgan's transformation lies in **quantum biology**, a growing field that

seeks to understand how quantum effects—those tiny, often counterintuitive processes that occur on a subatomic scale—might influence biological systems. The idea that quantum effects could influence the behaviour of biological matter offers a way to explain rapid, seemingly impossible transformations. For instance, quantum entanglement and tunnelling phenomena, where particles can exist in multiple states at once or even "pass through" barriers, could one day be harnessed to manipulate biological systems (Marais et al., 2018). While this is still speculative, it suggests that Morgan's shape-shifting could be akin to the hypothetical quantum changes we might one day trigger in biological organisms.

A similar idea can be applied to the molecular behaviour of materials. **Shape-shifting materials** that change their properties could one day be designed with properties derived from both natural and synthetic systems, allowing organisms or objects to alter their form at the microscopic level. This reconfiguring of molecular structures could mirror Morgan's magical abilities—transforming biological tissue into something as immovable and solid as stone before reverting to its original state.

2.5 From Ancient Magic to Modern Science

Through the lens of modern science, we can reinterpret Morgan le Fay's magical transformation from "man to stone" as an early exploration into the possibilities of material manipulation. In the world of **material science**, **bioengineering**, and **quantum biology**, scientists are already experimenting with concepts that resemble magical transformations, such as the ability to change the form, properties, and resilience of living and non-living matter. While Morgan's abilities may have seemed magical to medieval audiences, they may also represent a kind of proto-science—a precursor to ideas that we are only beginning to explore today.

3. The Forgotten Physician of the Middle Ages: Morgan le Fay and the Medical Arts

Far from being merely a literary enchantress or seductive antagonist, Morgan le Fay appears across medieval texts as a deeply learned figure—a healer, scientist, and educator whose abilities uncannily anticipate elements of modern medicine. Her portrayal demands a critical reevaluation, not only as a fictional figure but as an allegorical embodiment of female medical wisdom erased from dominant historical narratives. Through close reading and intertextual comparisons, Morgan's image is reclaimed as a forgotten pioneer of early scientific practice.

3.1 The Isle of Apples: Sovereignty, Nature, and Healing

Morgan's dominion over the Isle of Apples in *Vita Merlini* is not merely symbolic—it is functional, agricultural, and medicinal. The text introduces her as the leader of nine sisters and describes her in precise, technical terms:

She who is first of them is more skilled in the healing art than any other; she knows the use of all the herbs in healing as well as the art of curing sick bodies.
(Parry, n.d., p. 36).

The Isle becomes a metaphorical hospital of sorts—a utopia of healing governed by women. Morgan's expertise with herbs and the human body reflects the framework of medieval herbology, especially as found in *The Secrets of Albertus Magnus*, which categorizes plants by their therapeutic properties. Morgan's methods were not mere enchantments but aligned with empirically observed phenomena, functioning as proto-pharmacology.

3.2 Wound Healing and Surgery: Morgan as Trauma Physician

After the Battle of Camlan, Morgan takes the wounded King Arthur to Avalon. She asserts her ability to heal him if he remains under her care: “Health could be restored to him if he stayed with her for a long time and made use of her healing art”.(Parry, n.d., p. 37).

This act mirrors the use of *Potentilla* (Cinquefoil) for wound healing in *The Secrets of Albertus Magnus*, a plant known for its anti-inflammatory and antimicrobial properties—functions echoed in modern agents such as Silver Sulfadiazine and Epidermal Growth Factor (EGF). Morgan’s healing of Arthur marks her not only as a restorer of kingship but as a proto-surgeon, one whose botanical knowledge parallels the antiseptic and regenerative logic of contemporary medicine.

3.3 Sleep Induction and Anaesthesia: Sedation Before Science

In *Le Morte d’Arthur*, Morgan treats a knight who “waked never” for three days and nights: “She gave him such a drink that in three days and three nights he waked never, but slept”.(Malory, n.d., Book X, Chapter XXXVII).

This deliberate and extended sedation mirrors traditional soporifics such as Mephitis and Valerian root. Modern parallels include intravenous anaesthetics like Propofol and dissociatives like Ketamine, both of which, in sufficient dosage, induce medically managed unconsciousness. What Morgan practices is not witchcraft—it is pharmacokinetics avant la lettre.

3.4 Beauty as Science: Morgan's Dermatology

Morgan's beautification of her sisters is not merely cosmetic; it is calculated and pharmacological. As Geoffrey writes: "She gave to her sisters that which made them more beautiful than they had been before".(Parry, n.d., p. 38).

This is deeply aligned with medieval herbalism, which used Melissa, Vervain, and rose for skin renewal, hydration, and anti-aging. Morgan's techniques anticipate modern dermatological tools like Hyaluronic Acid, Retinoids, and Botox. Seen through this lens, her healing extends beyond disease—it addresses well-being, confidence, and social agency through aesthetic science.

3.5 Mental Health and Madness: Morgan the Psychiatrist

Perhaps the most striking instance of Morgan's healing is her treatment of mental illness in *Yvain* by Chrétien de Troyes. When a knight falls into madness, Morgan heals him: "She cured him with a certain ointment that she made and anointed him with; he recovered so completely that he never again showed any sign of madness".(Comfort, n.d., p. 79).

This resembles Coralwort (*Polygonum*), an herb noted in *The Secrets of Albertus Magnus* for its effect on disordered minds. Morgan's cure anticipates psychopharmacological interventions using SSRIs such as Fluoxetine (Prozac) and Sertraline (Zoloft), and antipsychotics like Olanzapine. The ointment is not fantasy—it is chemistry meeting psychiatry in its infancy.

6. Mathematics, Mentorship, and the Hidden Curriculum of Women

Morgan's intelligence is further validated by her role as a teacher. In *Vita Merlini*, Geoffrey includes a detail that scholars often overlook: "And men say that she has taught mathematics to

her sisters...".(Parry, n.d., p. 38).

To teach mathematics in the twelfth century was to operate within the quadrivium—arithmetic, geometry, music, and astronomy—the most advanced form of knowledge available to women only through unofficial or hidden networks. Her engagement with numerical logic further shatters the stereotype of Morgan as “irrational feminine chaos.” She is, in fact, the opposite: an orderly practitioner of natural law.

Morgan le Fay’s multifaceted healing across mental, physical, and aesthetic domains is not a fantastical embellishment but an allegory of women’s medical labor in the Middle Ages. Her interventions mirror the logic of both medieval medicine and modern pharmacology, suggesting a continuity of practice interrupted by patriarchal erasure. As modern science begins to rediscover the knowledge of “witches” and “wise women,” Morgan’s legacy becomes clearer—not as myth, but as a masked memory of female scientific genius.

4. Why Reimagining Morgan le Fay Matters: A Conclusion

Now that her story has been unfolded layer by layer—through wound care and sedation, through mental health and mathematics—we return to the question that underpins it all: *Why does the reimagination of Morgan le Fay matter?*

Let’s not shy away from the discomfort. Recasting a medieval sorceress as a proto-scientist raises eyebrows. It challenges the tidy boxes in which we place myth and science, reason and superstition, man and woman. Isn’t it dangerous, some might ask, to collapse these categories? But perhaps it is more dangerous *not to*. Because every time we consign a woman’s knowledge to fantasy, we reinforce a hierarchy of who gets to define what counts as *truth*.

There's an ugly double standard, centuries old and still breathing. When men pursued experimentation, transformation, and mastery over nature, they became alchemists, physicians, philosophers. When women did the same, they were burned or banished. Morgan le Fay is not the exception; she is the rule—made visible.

So let's ask: Why is Hildegard of Bingen honoured for her herbal wisdom, while Morgan is silenced as a fiction? Why do we celebrate Da Vinci's anatomical sketches, but not Morgan's precise, repeatable interventions in wound care, anaesthesia, and psychiatric healing? What does it say about us—about our epistemologies, our gendered frameworks of authority—that a woman doing science must first be stripped of her humanity or transformed into myth?

Even today, the resonance echoes. In Bengal and beyond, women in medicine still contend with erasure—not only from textbooks, but from policy, pay, and power. To reimagine Morgan is to pull back the curtain on this long lineage of forgotten healers and scientists who worked in the shadows while others took the spotlight.

If the Renaissance is the age of rediscovery, then Morgan le Fay was already there—centuries early, scripting her own revolution in a world that wasn't ready to listen. Reclaiming her is not about resurrecting a fantasy. It's about confronting the quiet violence of historical exclusion, and daring to imagine a different intellectual inheritance.

So let us end not with certainty, but with a question—one that resists easy closure: **What did she know that we've chosen to forget? And what might it mean if we finally choose to remember?**

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